# TP - Installation du Service WEB


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## Objectif :

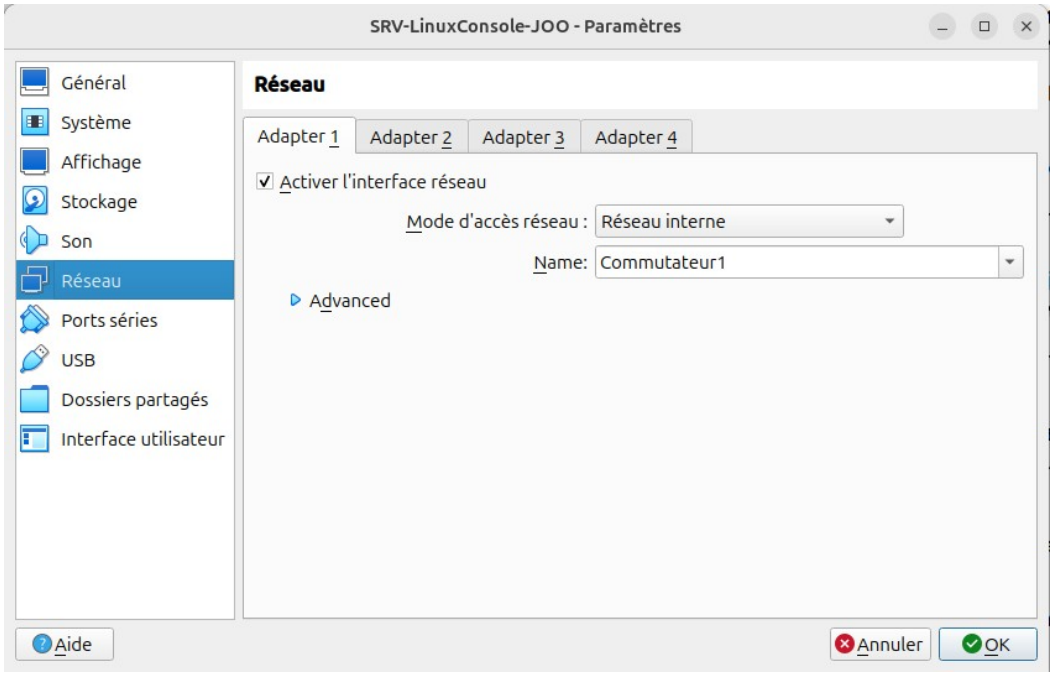
**L'objectif de ce TP est d'installer et de configurer un service WEB (HTTP) et de comprendre le fonctionnement du protocole HTTP. Vous allez utiliser un clone de votre machine virtuelle Linux Debian Serveur sans interface graphique. Ce service devra être intégré dans votre réseau interne.**

# I.Configuration de la machine

### Nommage du serveur




### Configuration des interfaces réseau du serveur ( Couche Physique )




# Configuration de l'interface réseau du serveur ( Couche Réseau )

```
root@debian-srv-WEB-YBA:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:ff:81:4d brd ff:ff:ff:ff:ff:ff
    altname enx080027ff814d
root@debian-srv-WEB-YBA:~#
```

```
C:\Users\sio>ping 192.168.56.3

Envoi d'une requête 'Ping'  192.168.56.3 avec 32 octets de données :
Réponse de 192.168.56.3 : octets=32 temps<1ms TTL=64
Réponse de 192.168.56.3 : octets=32 temps<1ms TTL=64
Réponse de 192.168.56.3 : octets=32 temps<1ms TTL=64
Réponse de 192.168.56.3 : octets=32 temps<1ms TTL=64

Statistiques Ping pour 192.168.56.3:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 0ms, Maximum = 0ms, Moyenne = 0ms
```

```
root@debian-srv-WEB-YBA:/# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=6.28 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=5.01 ms
^C
--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 5.007/5.644/6.281/0.637 ms
root@debian-srv-WEB-YBA:/# ping www.google.fr
PING forcesafesearch.google.com (216.239.38.120) 56(84) bytes of data.
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=1 ttl=110 time=6.58 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=2 ttl=110 time=5.54 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=3 ttl=110 time=5.58 ms
^C
--- forcesafesearch.google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 5.543/5.902/6.584/0.482 ms
root@debian-srv-WEB-YBA:/# ping 192.168.56.250
PING 192.168.56.250 (192.168.56.250) 56(84) bytes of data.
64 bytes from 192.168.56.250: icmp_seq=1 ttl=64 time=2.34 ms
64 bytes from 192.168.56.250: icmp_seq=2 ttl=64 time=0.690 ms

--- 192.168.56.250 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
^Crtt min/avg/max/mdev = 0.690/1.515/2.340/0.825 ms
```

```
root@debian-srv-WEB-YBA:/# ping 172.16.0.51
PING 172.16.0.51 (172.16.0.51) 56(84) bytes of data.
64 bytes from 172.16.0.51: icmp_seq=1 ttl=62 time=2.80 ms
64 bytes from 172.16.0.51: icmp_seq=2 ttl=62 time=1.55 ms
^C
--- 172.16.0.51 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.552/2.175/2.799/0.623 ms
```

## Étape 1 : Installation du logiciel serveur HTTP.

```
make-ssl-cert: You may want to fix your /etc/hosts and/or DNS setup and run
make-ssl-cert: 'make-ssl-cert generate-default-snakeoil --force-overwrite'
make-ssl-cert: again.
Paramétrage de librtmp1:amd64 (2.4+20151223.gitfa8646d.1-2+b5) ...
Paramétrage de libsasl2-2:amd64 (2.1.28+dfsg1-9) ...
Paramétrage de libnghttp3-9:amd64 (1.8.0-1) ...
Paramétrage de libapr1t64:amd64 (1.7.5-1) ...
Paramétrage de liblua5.4-0:amd64 (5.4.7-1+b2) ...
Paramétrage de apache2-data (2.4.66-1~deb13u1) ...
Paramétrage de libssh2-1t64:amd64 (1.11.1-1) ...
Paramétrage de libldap2:amd64 (2.6.10+dfsg-1) ...
Paramétrage de libaprutil1t64:amd64 (1.6.3-3+b1) ...
Paramétrage de libcurl4t64:amd64 (8.14.1-2+deb13u2) ...
Paramétrage de libaprutil1-ldap:amd64 (1.6.3-3+b1) ...
Paramétrage de libaprutil1-dbd-sqlite3:amd64 (1.6.3-3+b1) ...
Paramétrage de apache2-utils (2.4.66-1~deb13u1) ...
Paramétrage de apache2-bin (2.4.66-1~deb13u1) ...
Paramétrage de apache2 (2.4.66-1~deb13u1) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' → '/usr/lib/systemd/system/apache2.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service' → '/usr/lib/systemd/system/apache-htcacheclean.service'.
Traitement des actions différées (« triggers ») pour man-db (2.13.1-1) ...
Traitement des actions différées (« triggers ») pour libc-bin (2.41-12+deb13u1) ...
```

```
root@debian-srv-WEB-YBA:~# apt install net-tools
net-tools est déjà la version la plus récente (2.10-1.3).
Sommaire :
  Mise à niveau de : 0. Installation de : 0Supprimé : 0. Non mis à jour : 0
```

## Étape 2 : Contrôle du bon fonctionnement du service

## Vérification du fonctionnement du service

```
root@debian-srv-WEB-YBA:~# systemctl -t service --state=running
  UNIT                        LOAD   ACTIVE SUB     DESCRIPTION
  apache-htcacheclean.service loaded active running Disk Cache Cleaning Daemon for Apache HTTP Server
  apache2.service             loaded active running The Apache HTTP Server
  cron.service                loaded active running Regular background program processing daemon
  dbus.service                loaded active running D-Bus System Message Bus
  getty@tty1.service          loaded active running Getty on tty1
  ssh.service                 loaded active running OpenBSD Secure Shell server
  systemd-journald.service    loaded active running Journal Service
  systemd-logind.service      loaded active running User Login Management
  systemd-timesyncd.service   loaded active running Network Time Synchronization
  systemd-udevd.service       loaded active running Rule-based Manager for Device Events and Files
  user@0.service              loaded active running User Manager for UID 0

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
        SUB    → The low-level unit activation state, values depend on unit type.

11 loaded units listed.
```

```
root@debian-srv-WEB-YBA:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-09 14:16:36 CET; 5min ago
 Invocation: 541da180b6d348df91a935ff7358b4b6
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1347 (apache2)
      Tasks: 55 (limit: 1093)
     Memory: 5.5M (peak: 5.7M)
        CPU: 56ms
     CGroup: /system.slice/apache2.service
             ├─1347 /usr/sbin/apache2 -k start
             ├─1350 /usr/sbin/apache2 -k start
             └─1351 /usr/sbin/apache2 -k start

mars 09 14:16:36 debian-srv-WEB-YBA systemd[1]: Starting apache2.service - The Apache HTTP Server...
mars 09 14:16:36 debian-srv-WEB-YBA apachectl[1346]: AH00557: apache2: apr_sockaddr_info_get() failed for debian-srv-WEB-YBA
mars 09 14:16:36 debian-srv-WEB-YBA apachectl[1346]: AH00558: apache2: Could not reliably determine the server's fully qualified
mars 09 14:16:36 debian-srv-WEB-YBA systemd[1]: Started apache2.service - The Apache HTTP Server.
```

```
root@debian-srv-WEB-YBA:~# netstat -tulpn | grep -i apache
tcp6       0      0 :::80                   :::*                    LISTEN      1347/apache2
```

# Vérification des journaux système

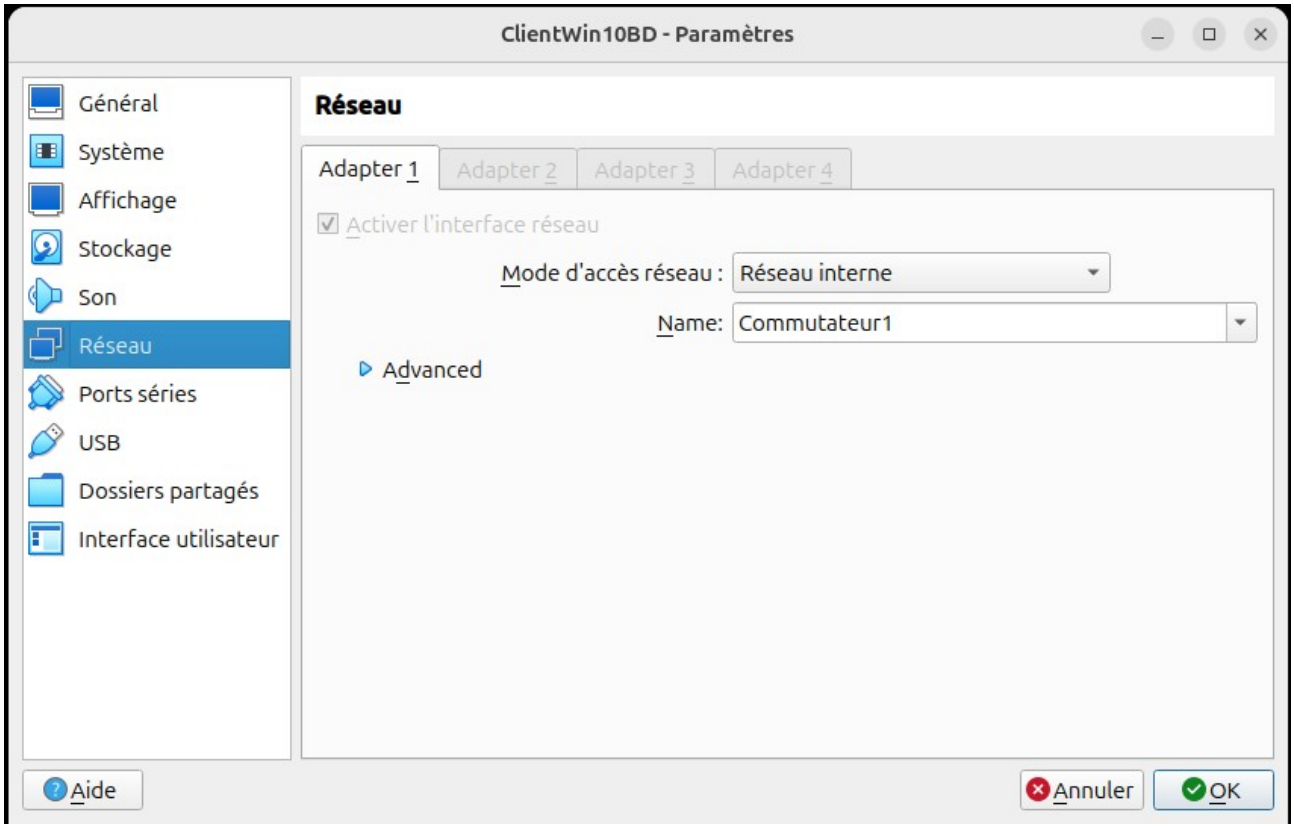
```
janv. 19 11:20:03 debian-srv-linux-yA kernel: Linux version 6.12.63+deb13-amd64 (debian-kernel@lists.debian.org) (x86_64-linux-gnu-gcc-14 (Debian 14.2.0-19)
janv. 19 11:20:03 debian-srv-linux-yA kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.63+deb13-amd64 root=UUID=d94fa6a3-41dc-4ba2-88eb-b6faddc3fcf7 ro qu
janv. 19 11:20:03 debian-srv-linux-yA kernel: BIOS-provided physical RAM map:
janv. 19 11:20:03 debian-srv-linux-yA kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
janv. 19 11:20:03 debian-srv-linux-yA kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
janv. 19 11:20:03 debian-srv-linux-yA kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
janv. 19 11:20:03 debian-srv-linux-yA kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000dffeffff] usable
janv. 19 11:20:03 debian-srv-linux-yA kernel: BIOS-e820: [mem 0x00000000dfff0000-0x00000000dfffffff] ACPI data
janv. 19 11:20:03 debian-srv-linux-yA kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
janv. 19 11:20:03 debian-srv-linux-yA kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
janv. 19 11:20:03 debian-srv-linux-yA kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
janv. 19 11:20:03 debian-srv-linux-yA kernel: NX (Execute Disable) protection: active
janv. 19 11:20:03 debian-srv-linux-yA kernel: APIC: Static calls initialized
janv. 19 11:20:03 debian-srv-linux-yA kernel: SMBIOS 2.5 present.
janv. 19 11:20:03 debian-srv-linux-yA kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
janv. 19 11:20:03 debian-srv-linux-yA kernel: DMI: Memory slots populated: 0/0
janv. 19 11:20:03 debian-srv-linux-yA kernel: Hypervisor detected: KVM
janv. 19 11:20:03 debian-srv-linux-yA kernel: last_pfn = 0x40000 max_arch_pfn = 0x400000000
janv. 19 11:20:03 debian-srv-linux-yA kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
janv. 19 11:20:03 debian-srv-linux-yA kernel: kvm-clock: using sched offset of 9110261771 cycles
janv. 19 11:20:03 debian-srv-linux-yA kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 881590591483 ns
janv. 19 11:20:03 debian-srv-linux-yA kernel: tsc: Detected 3092.972 MHz processor
janv. 19 11:20:03 debian-srv-linux-yA kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
janv. 19 11:20:03 debian-srv-linux-yA kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
janv. 19 11:20:03 debian-srv-linux-yA kernel: last_pfn = 0x40000 max_arch_pfn = 0x400000000
janv. 19 11:20:03 debian-srv-linux-yA kernel: MTRRs disabled by BIOS
janv. 19 11:20:03 debian-srv-linux-yA kernel: x86/PAT: Configuration [0-7]: WB  WC  UC- UC  WB  WP  UC- WT
janv. 19 11:20:03 debian-srv-linux-yA kernel: found SMP MP-table at [mem 0x0009fff0-0x0009ffff]
janv. 19 11:20:03 debian-srv-linux-yA kernel: RAMDISK: [mem 0x31df7000-0x34ef2fff]
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: Early table checksum verification disabled
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: RSDP 0x00000000000E0000 000024 (v02 VBOX  )
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: XSDT 0x00000000DFFF0030 00003C (v01 VBOX   VBOXXSDT 00000001 ASL  00000061)
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: FACP 0x00000000DFFF00F0 0000F4 (v04 VBOX   VBOXFACP 00000001 ASL  00000061)
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: DSDT 0x00000000DFFF0610 002353 (v02 VBOX   VBOXBIOS 00000002 INTL 20230628)
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: FACS 0x00000000DFFF0200 000040
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: FACS 0x00000000DFFF0200 000040
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: APIC 0x00000000DFFF0240 000054 (v02 VBOX   VBOXAPIC 00000001 ASL  00000061)
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: SSDT 0x00000000DFFF02A0 00036C (v01 VBOX   VBOXCPUT 00000002 INTL 20230628)
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: Reserving FACP table memory at [mem 0x3fff00f0-0x3fff01e3]
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: Reserving DSDT table memory at [mem 0x3fff0610-0x3fff2962]
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: Reserving FACS table memory at [mem 0x3fff0200-0x3fff023f]
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: Reserving FACS table memory at [mem 0x3fff0200-0x3fff023f]
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: Reserving APIC table memory at [mem 0x3fff0240-0x3fff0293]
janv. 19 11:20:03 debian-srv-linux-yA kernel: ACPI: Reserving SSDT table memory at [mem 0x3fff02a0-0x3fff060b]
janv. 19 11:20:03 debian-srv-linux-yA kernel: No NUMA configuration found
janv. 19 11:20:03 debian-srv-linux-yA kernel: Faking a node at [mem 0x0000000000000000-0x00000000dfffffff]
janv. 19 11:20:03 debian-srv-linux-yA kernel: NODE_DATA(0) allocated [mem 0x3ffc5680-0x3ffeffff]
janv. 19 11:20:03 debian-srv-linux-yA kernel: Zone ranges:
janv. 19 11:20:03 debian-srv-linux-yA kernel:   DMA      [mem 0x0000000000001000-0x0000000000ffffff]
```

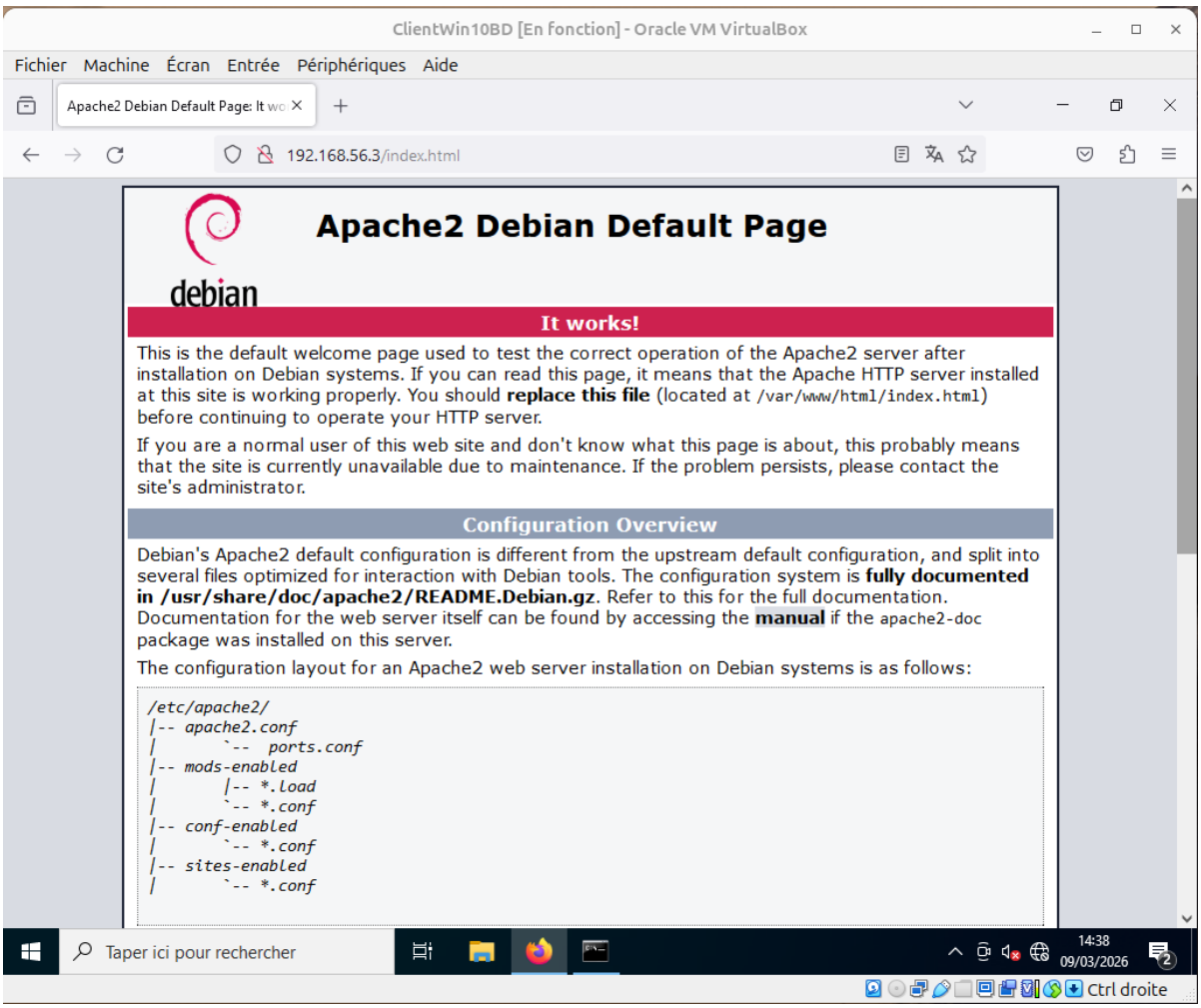
```
root@debian-srv-WEB-YBA:~# journalctl -u apache2
mars 09 14:16:36 debian-srv-WEB-YBA systemd[1]: Starting apache2.service - The Apache HTTP Server...
mars 09 14:16:36 debian-srv-WEB-YBA apachectl[1346]: AH00557: apache2: apr_sockaddr_info_get() failed for debian-srv-WEB-YBA
mars 09 14:16:36 debian-srv-WEB-YBA apachectl[1346]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.0.1.
mars 09 14:16:36 debian-srv-WEB-YBA systemd[1]: Started apache2.service - The Apache HTTP Server.
```

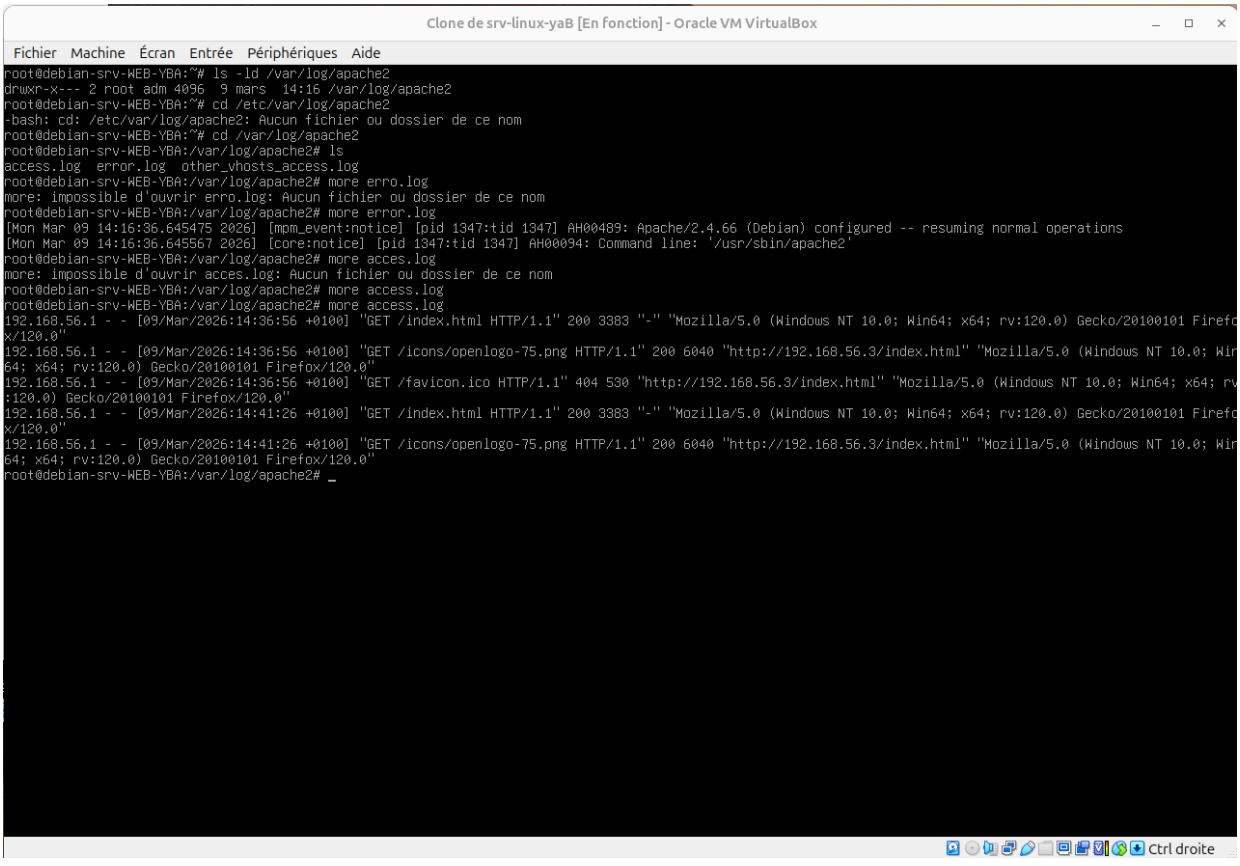
## Vérification des journaux applicatifs

```
root@debian-srv-WEB-YBA:/var/log/apache2# ls
access.log  error.log  other_vhosts_access.log
root@debian-srv-WEB-YBA:/var/log/apache2# more erro.log
more: impossible d'ouvrir erro.log: Aucun fichier ou dossier de ce nom
root@debian-srv-WEB-YBA:/var/log/apache2# more error.log
[Mon Mar 09 14:16:36.645475 2026] [mpm_event:notice] [pid 1347:tid 1347] AH00489: Apache/2.4.66 (Debian) configured -- resuming normal operations
[Mon Mar 09 14:16:36.645567 2026] [core:notice] [pid 1347:tid 1347] AH00094: Command line: '/usr/sbin/apache2'
root@debian-srv-WEB-YBA:/var/log/apache2# more acces.log
more: impossible d'ouvrir acces.log: Aucun fichier ou dossier de ce nom
root@debian-srv-WEB-YBA:/var/log/apache2# more access.log
```

## Étape 3 : Utilisation du service Web
## Principe de fonctionnement du service Web
## Test depuis une machine cliente




ClientWin10BD [En fonction] - Oracle VM VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
Apache2 Debian Default Page: It wo
192.168.56.3/index.html
Apache2 Debian Default Page
debian
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Debian systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf
Taper ici pour rechercher
14:38
09/03/2026
Ctrl droite


Clone de srv-linux-yaB [En fonction] - Oracle VM VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
root@debian-srv-WEB-YBA:~# ls -ld /var/log/apache2
drwxr-x--- 2 root adm 4096 9 mars 14:16 /var/log/apache2
root@debian-srv-WEB-YBA:~# cd /etc/var/log/apache2
-bash: cd: /etc/var/log/apache2: Aucun fichier ou dossier de ce nom
root@debian-srv-WEB-YBA:~# cd /var/log/apache2
root@debian-srv-WEB-YBA:/var/log/apache2# ls
access.log error.log other_vhosts_access.log
root@debian-srv-WEB-YBA:/var/log/apache2# more erro.log
more: impossible d'ouvrir erro.log: Aucun fichier ou dossier de ce nom
root@debian-srv-WEB-YBA:/var/log/apache2# more error.log
[Mon Mar 09 14:16:36.645475 2026] [mpm_event:notice] [pid 1347:tid 1347] AH00489: Apache/2.4.66 (Debian) configured -- resuming normal operations
[Mon Mar 09 14:16:36.645567 2026] [core:notice] [pid 1347:tid 1347] AH00094: Command line: '/usr/sbin/apache2'
root@debian-srv-WEB-YBA:/var/log/apache2# more acces.log
more: impossible d'ouvrir acces.log: Aucun fichier ou dossier de ce nom
root@debian-srv-WEB-YBA:/var/log/apache2# more access.log
root@debian-srv-WEB-YBA:/var/log/apache2# more access.log
192.168.56.1 - - [09/Mar/2026:14:36:56 +0100] "GET /index.html HTTP/1.1" 200 3383 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:120.0) Gecko/20100101 Firefox/120.0"
192.168.56.1 - - [09/Mar/2026:14:36:56 +0100] "GET /icons/openlogo-75.png HTTP/1.1" 200 6040 "http://192.168.56.3/index.html" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:120.0) Gecko/20100101 Firefox/120.0"
192.168.56.1 - - [09/Mar/2026:14:36:56 +0100] "GET /favicon.ico HTTP/1.1" 404 530 "http://192.168.56.3/index.html" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:120.0) Gecko/20100101 Firefox/120.0"
192.168.56.1 - - [09/Mar/2026:14:41:26 +0100] "GET /index.html HTTP/1.1" 200 3383 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:120.0) Gecko/20100101 Firefox/120.0"
192.168.56.1 - - [09/Mar/2026:14:41:26 +0100] "GET /icons/openlogo-75.png HTTP/1.1" 200 6040 "http://192.168.56.3/index.html" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:120.0) Gecko/20100101 Firefox/120.0"
root@debian-srv-WEB-YBA:/var/log/apache2#
Ctrl droite